AI-driven prediction of completion time and errors in the Advanced Training in Laparoscopic Suturing (ATLAS) needle handling task: One step closer to automated surgical skill assessment.

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Abstract

The Advanced Training in Laparoscopic Suturing is a proficiency-based curriculum of 6 structured tasks. In the needle handling task, participants maneuver a needle through 6 standardized holes on a circular platform. Performance (completion time and errors) is currently evaluated in person or through manual video review. This study explored the potential of artificial intelligence models to automate the assessment of this task by predicting task duration and detecting needle drop errors. A retrospective review was conducted of Advanced Training in Laparoscopic Suturing needle handling task videos collected from 2 tertiary centers. Two complementary artificial intelligence models were developed. First, videos were annotated across 10 distinct phases. A deep expandable three-dimensional convolutional network combined with hybrid adaptive k-nearest neighbors and smoothed moving average and exponential moving average was trained for phase segmentation and duration prediction. Second, a vision transformer model was trained to detect needle drop errors by classifying frame segments based on needle visibility. Phase segmentation accuracy improved from 82.06% ± 0.84% to 89.67% ± 1.27%, with the highest accuracy reaching 90.56% and an F1-score of 86.90% using the hybrid k-nearest neighbors and smoothed moving average model. The predicted task duration error had a mean error of 0.84%. The vision transformer model achieved a 95.16% classification accuracy on validation frames and detected 66.6% of needle drops >2 seconds and 63.6% of needle drops >5 seconds in test videos. Artificial intelligence-based models exhibited high and moderate accuracy for task duration prediction and needle drop error, respectively, offering scalable solutions for objective surgical assessments.

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